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Machine Learning Cheat Sheet



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Types of Machine Learning

SUPERVISED LEARNING

Regression

- Linear
- Polynomial
- Ridge ,
- Lasso ,
- Decision Tree,
- Random Forest,
- SVR
- XGBoost

Classification

- Logistic
- Decision Tree
- Random Forest
- SVM
- k-NN
- Naïve Bayes
- XGBoost
- Neural Networks

UNSUPERVISED LEARNING

Clustering

- k-Means,
- Hierarchical
- DBSCAN,
- GMM
- Mean Shift

Dimensionality

- k-Means,
- Hierarchical
- DBSCAN,
- GMM
- Mean Shift

REINFORCEMENT LEARNING

Reward-based learning

- Q-Learning
- Deep Q-Networks (DQN)
- SARSA (State-Action-Reward-State-Action)
- Policy Gradient Methods
- Actor-Critic Methods
- Proximal Policy Optimization (PPO)
- Trust Region Policy Optimization (TRPO)
- Monte Carlo Methods



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Data Preprocessing

`df.isnull().sum()`: Check missing values

`df.dropna()`: Remove missing values

`df.fillna(value)`: Fill missing values

`StandardScaler()`:

Standardization (mean=0, std=1)

`MinMaxScaler()`: Normalization (range [0,1])

`LabelEncoder()`: Convert categorical labels to numbers

`OneHotEncoder()`: Convert categorical features into dummy variables



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